Kernix 0.11.1 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

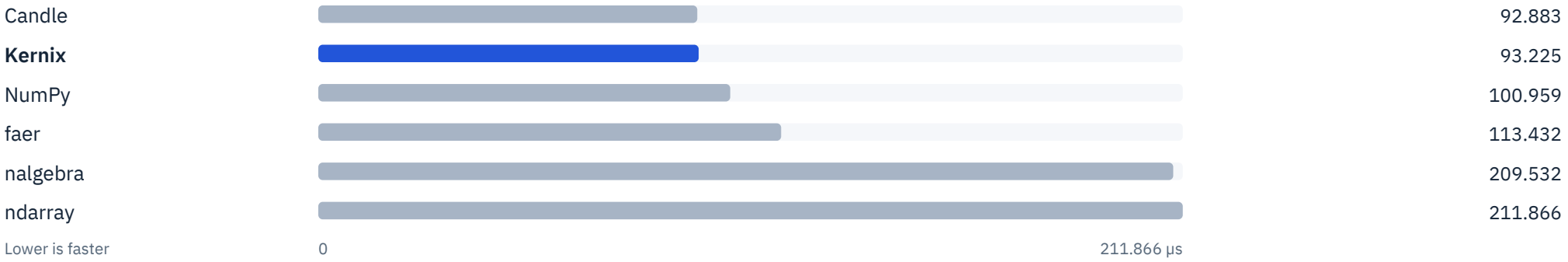
35 workloads

119 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



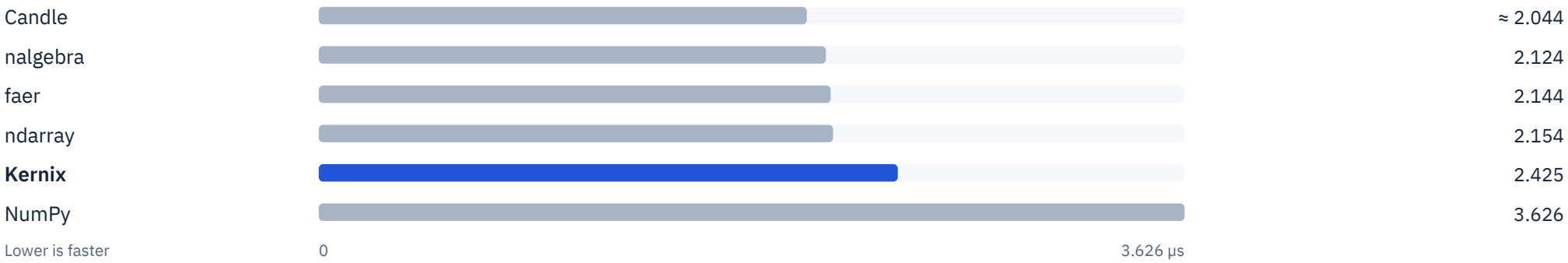
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

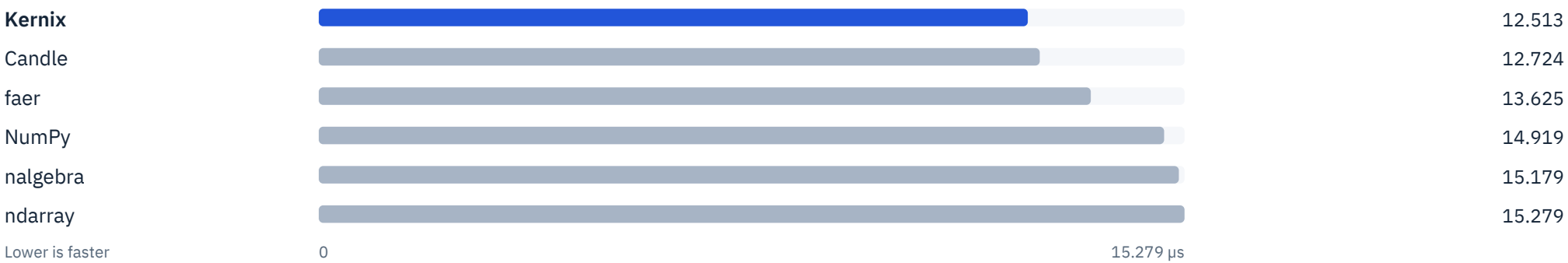
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



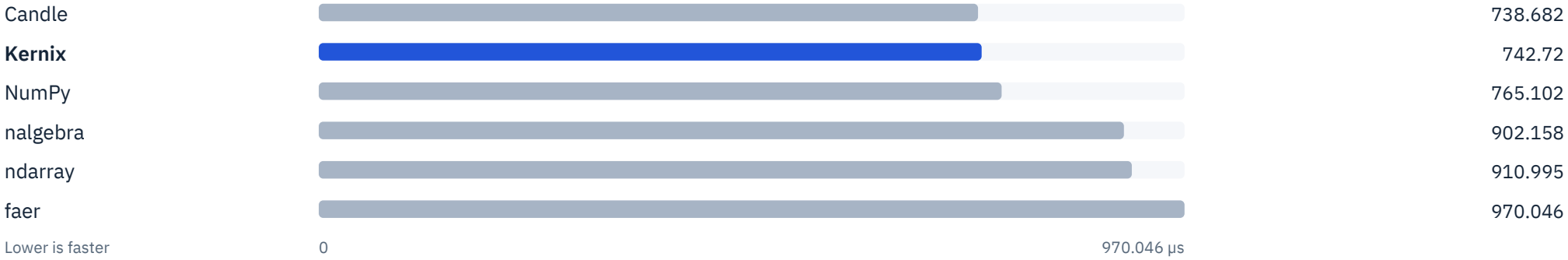
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

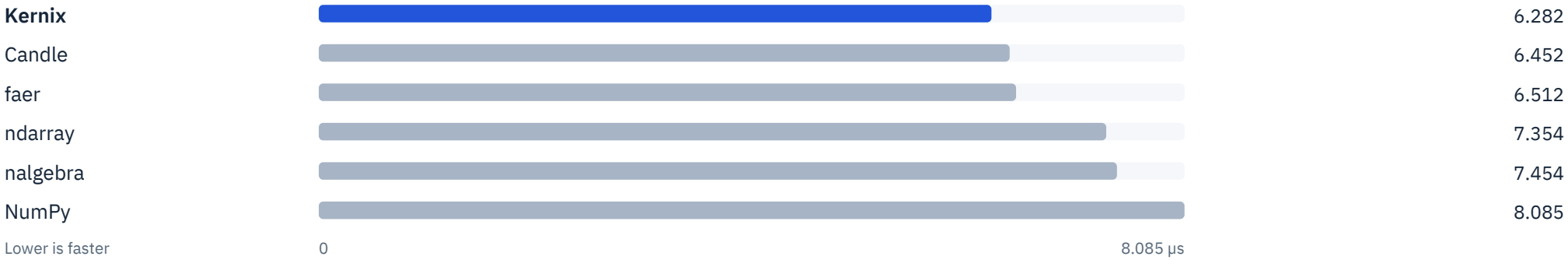
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



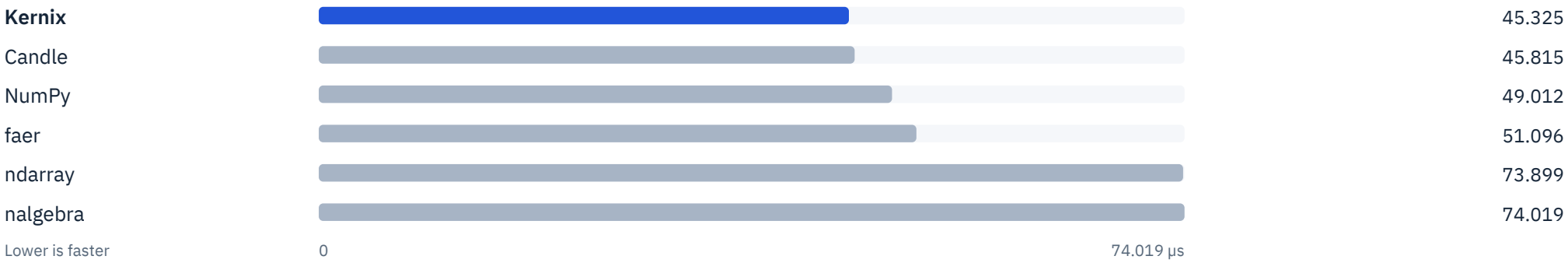
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

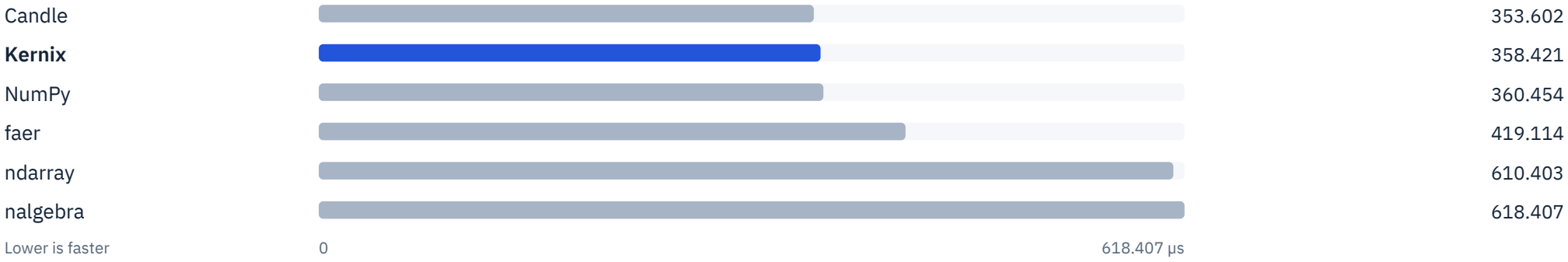
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



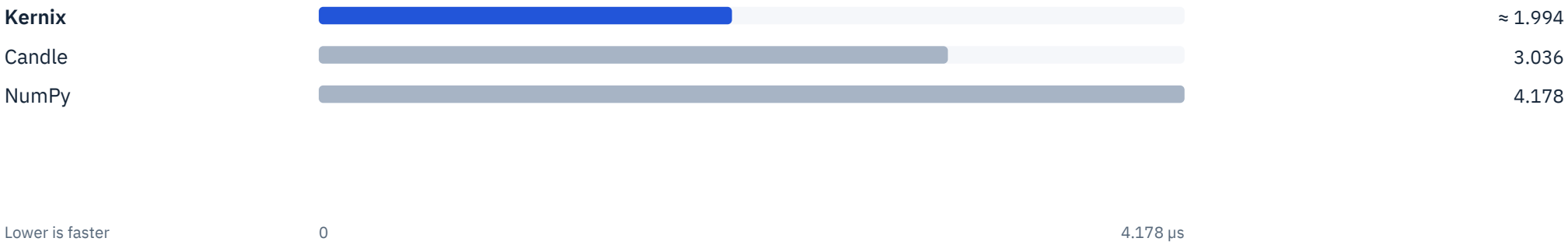
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

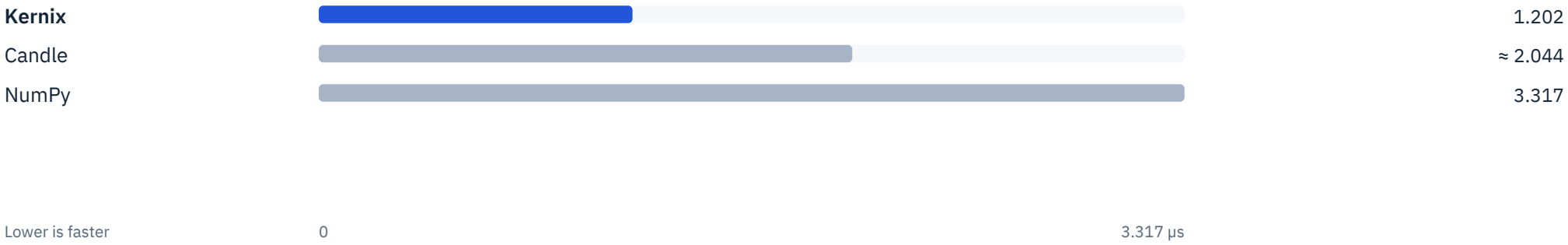
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

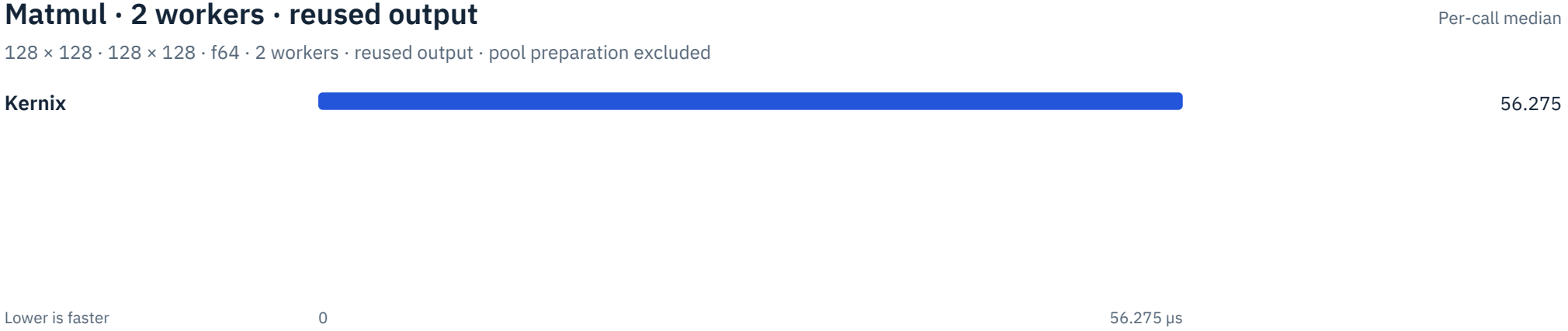
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

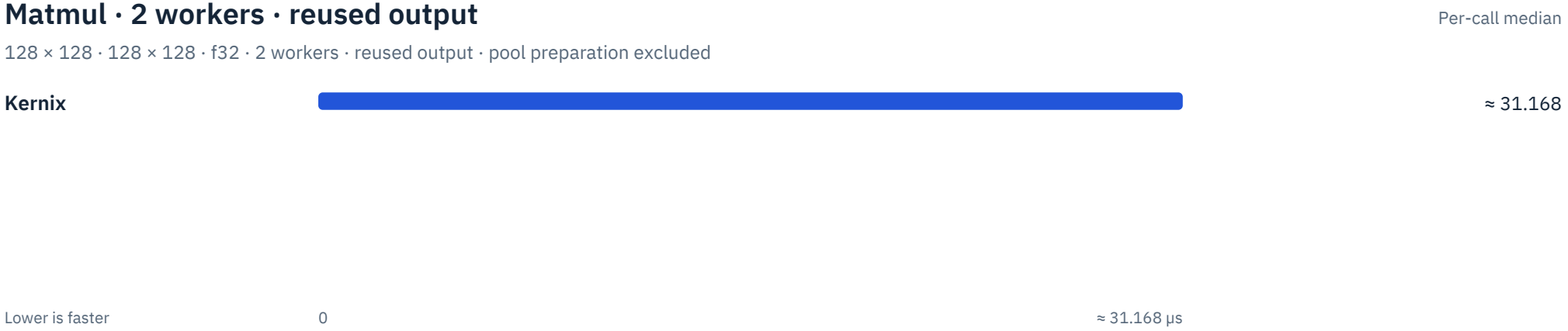
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



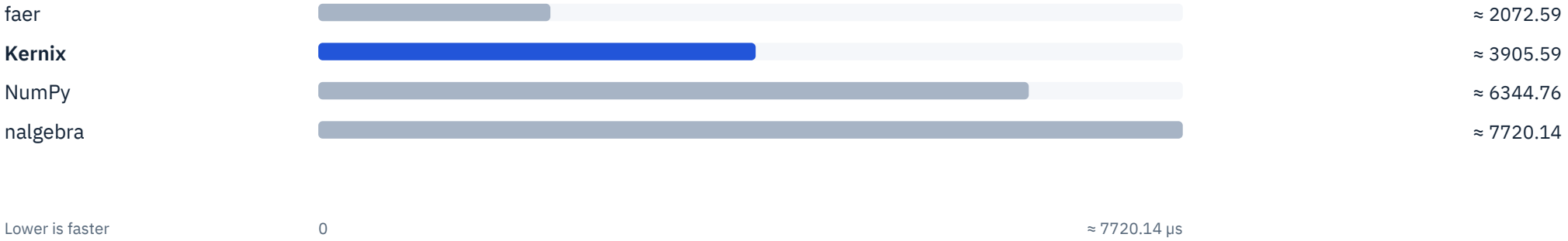
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

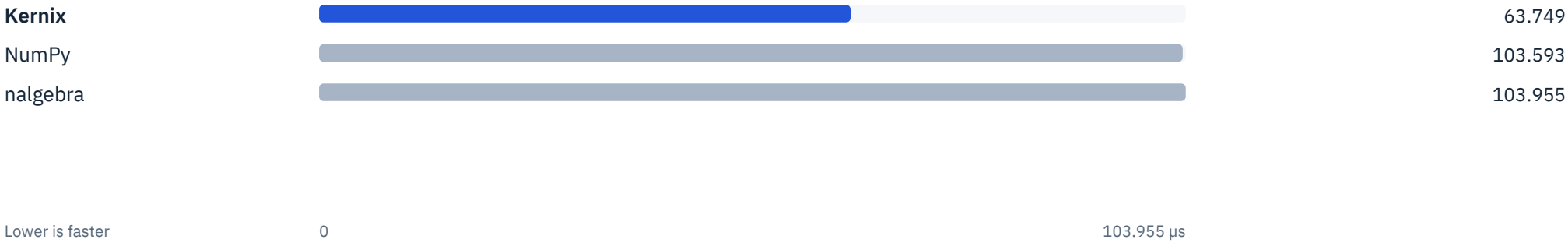
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



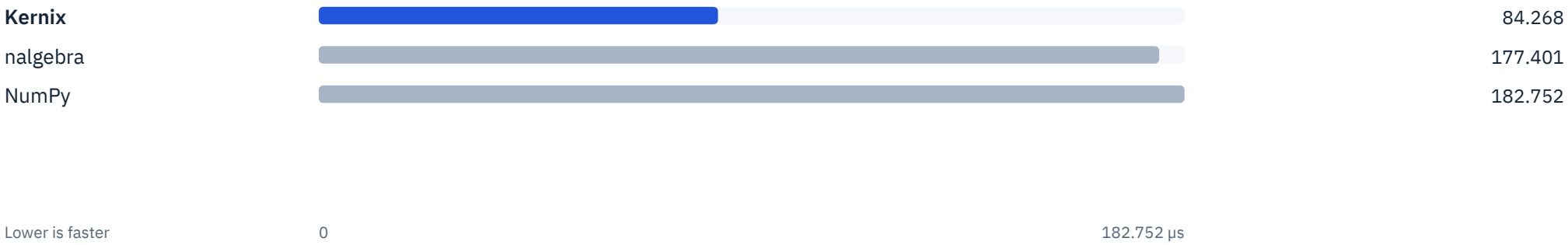
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

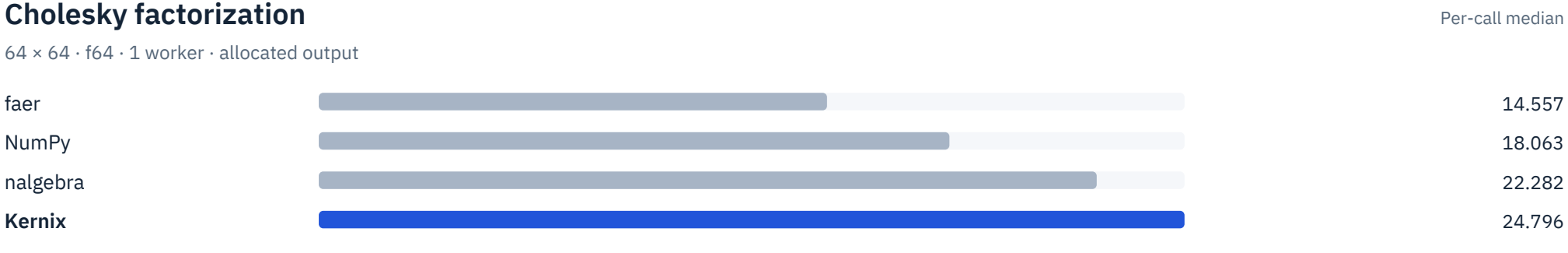
Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

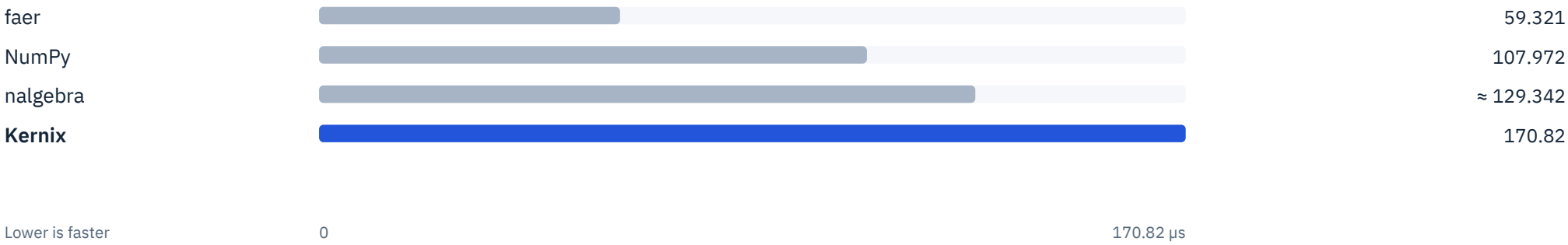


Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

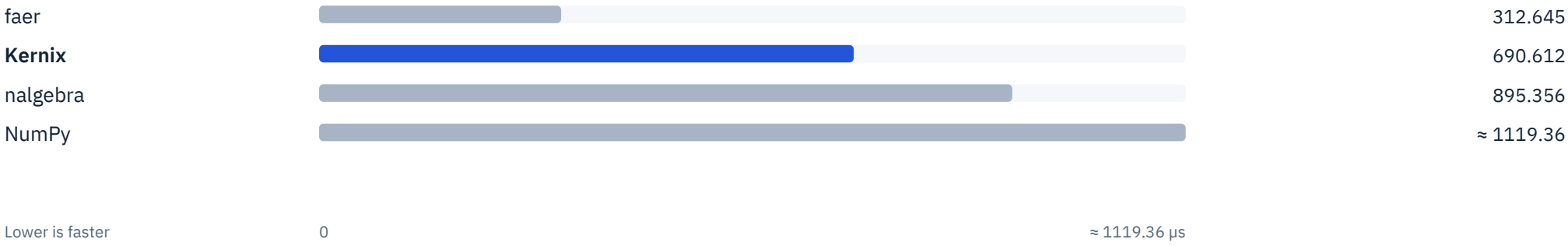
Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



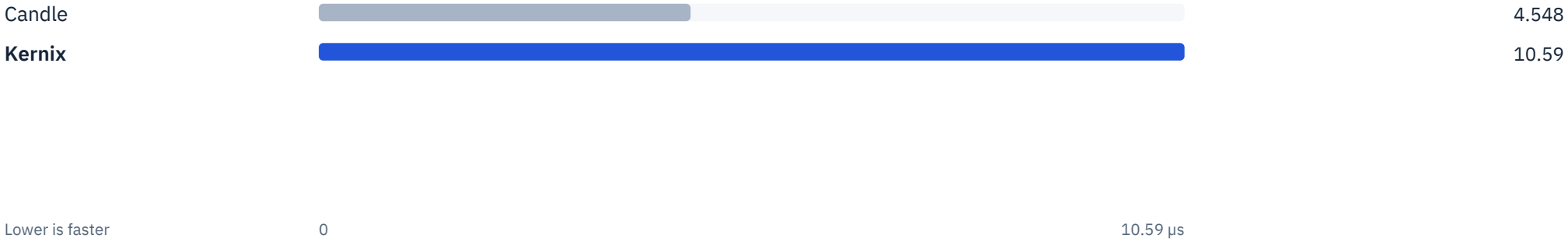
Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



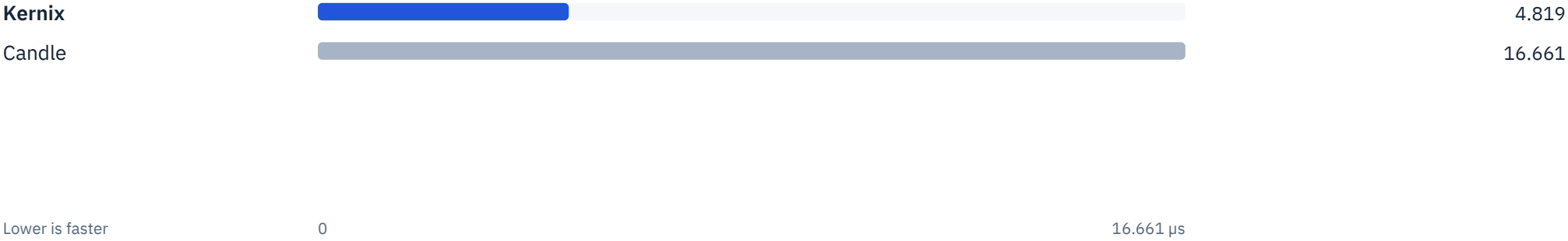
Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



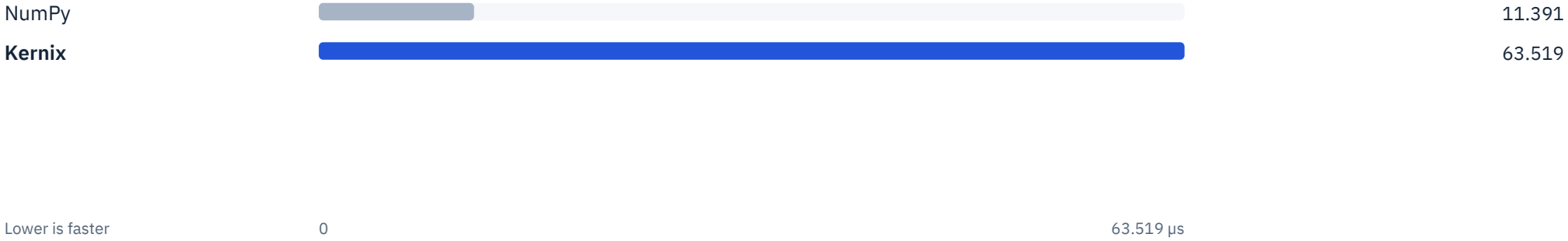
Kernix 0.11.1 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



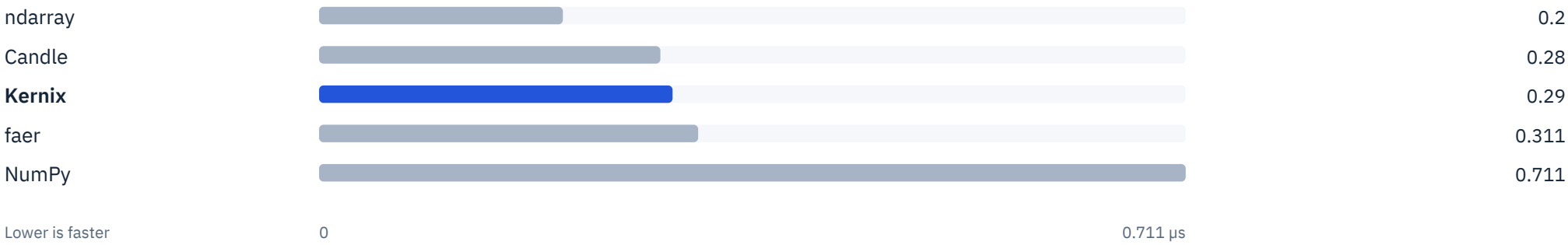
Kernix 0.11.1 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0041407867494825	1 – 1.0045379537953796

Matrix multiplication

128 × 128 · 128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0099791109716103	1.0052981990439502 – 1.0114145589878802

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.000220482857458	0.9982380795066622 – 1.0015236833388539

Matrix multiplication

256 × 256 · 256 × 256 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.006558129554634	1.0040813729657436 – 1.0097556013671134

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0022646283129366	1.0001415744780857 – 1.0036768129396727

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9219211161792327	0.9046610169491526 – 0.9351600651112317

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0073907695598696	1.0036293584230918 – 1.0111569141817014

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0028731220528566	0.9951774102038144 – 1.0099328267811916

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Observed time reduction

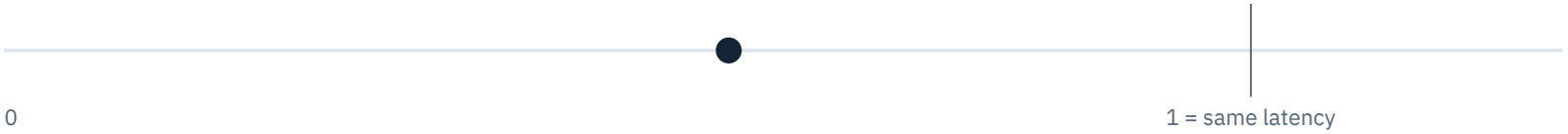


CURRENT / BASELINE	95% INTERVAL
0.659749670619236	0.6589557171183079 – 0.6666666666666666

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.5811209439528023	0.5733855185909981 – 0.5885518590998043

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9937565403473292	0.9928150613248039 – 1.0013695787507781

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.994797745689799	0.9879273217853035 – 1.0047720257365078

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9983466021802162	0.9966300720587722 – 1.002325631779009

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0052951143744704	0.9939748392112516 – 1.0154761476794765

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.009035905294692	1.0026501946355793 – 1.0426416803806298

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9993552131020697	0.9974357330598115 – 1.0067443286327409

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	103043	103.043
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	94016	94.016
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	49052	49.052
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	45635	45.635
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2043.9999999999998	2.0439999999999998
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2154	2.154
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2124	2.124
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2144	2.144
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	3626	3.626
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2425	2.425

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15279	15.279
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	14919	14.919
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12724	12.724
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13625	13.625
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	15179	15.179
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12513	12.513
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100959	100.959
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	211866	211.866
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	93225	93.225
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	113432	113.432

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	209532	209.532
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	92883	92.883
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	765102	765.102
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	738682	738.682
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	902158	902.158
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	910995	910.995
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	970046	970.046
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	742720	742.72
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7354	7.354
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8085	8.085

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6512	6.512
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6282	6.282
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6452	6.452
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7454	7.454
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	74019	74.019
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	51096	51.096
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	45815	45.815
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	73899	73.899
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	45325	45.325
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	49012	49.012

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	618407	618.407
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	360454	360.454
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	610403	610.403
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	419114	419.114
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	358421	358.421
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	353602	353.602
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3036	3.036
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4178	4.178
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1993.9999999999998	1.9939999999999998
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2043.9999999999998	2.0439999999999998

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	3317	3.317
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1202	1.202
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	15688.999999999998	15.688999999999998
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	56275	56.275
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	401331	401.331
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	31167.999999999996	31.167999999999996
Cholesky factorization / nalgebra 512 × 512 · f64	7720137	7720.137
Cholesky factorization / NumPy 512 × 512 · f64	6344763	6344.763
Cholesky factorization / faer 512 × 512 · f64	2072588.0000000002	2072.5880000000002
Cholesky factorization / Kernix 512 × 512 · f64	3905588	3905.588

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	42590	42.59
Column-pivoted QR / nalgebra 96 × 32 · f64	78347	78.347
Column-pivoted QR / Kernix 96 × 32 · f64	78938	78.938
Column-pivoted QR / nalgebra 256 × 64 · f64	749562	749.562
Column-pivoted QR / faer 256 × 64 · f64	383948	383.948
Column-pivoted QR / Kernix 256 × 64 · f64	810206	810.206
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103955	103.955
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	103593	103.593
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	63749	63.749
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	182752	182.752

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	84268	84.268
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	177401	177.401
Thin QR / faer 96 × 32 · f64	55474	55.474
Thin QR / NumPy 96 × 32 · f64	69430	69.43
Thin QR / Kernix 96 × 32 · f64	66223	66.223
Thin QR / nalgebra 96 × 32 · f64	58550	58.55
Thin QR / faer 256 × 64 · f64	423843	423.843
Thin QR / nalgebra 256 × 64 · f64	566359	566.359
Thin QR / Kernix 256 × 64 · f64	758811	758.811
Thin QR / NumPy 256 × 64 · f64	537225	537.225

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	24796	24.796
Cholesky factorization / faer 64 × 64 · f64	14557	14.557
Cholesky factorization / NumPy 64 × 64 · f64	18063	18.063
Cholesky factorization / nalgebra 64 × 64 · f64	22282	22.282
Cholesky factorization / NumPy 128 × 128 · f64	107972	107.972
Cholesky factorization / nalgebra 128 × 128 · f64	129342.00000000001	129.34200000000001
Cholesky factorization / faer 128 × 128 · f64	59321	59.321
Cholesky factorization / Kernix 128 × 128 · f64	170820	170.82
Cholesky factorization / NumPy 256 × 256 · f64	1119364	1119.364
Cholesky factorization / faer 256 × 256 · f64	312645	312.645

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	895356	895.356
Cholesky factorization / Kernix 256 × 256 · f64	690612	690.612
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4548	4.548
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10590	10.59
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	5511	5.511
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3416	3.416
Model inference / Candle batched normalized/sliced MLP · f64	4298	4.298
Model inference / Kernix batched normalized/sliced MLP · f64	2966	2.966
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3848.0000000000005	3.8480000000000005
Model recording + forward / Candle batched normalized/sliced MLP · f64	6191	6.191

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	4819	4.819
Model backward / Candle prepared graph → fresh gradients · f64	16661	16.661
Parameter update / Kernix transactional SGD, fixed gradients · f64	781	0.781
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63519	63.519
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	11391	11.391
Row softmax / Candle 256 × 64 · f64	84839	84.839
Row softmax / ndarray 256 × 64 · f64	80751	80.751
Row softmax / Kernix 256 × 64 · f64	94146	94.146
Row softmax / NumPy 256 × 64 · f64	119814	119.814
Create zeros / faer 64 × 64 · f32	311	0.311

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	280	0.28
Create zeros / NumPy 64 × 64 · f32	711	0.711
Create zeros / Kernix 64 × 64 · f32	290	0.29
Create zeros / ndarray 64 × 64 · f32	200	0.2
Create ones / Candle 64 × 64 · f32	441	0.441
Create ones / ndarray 64 × 64 · f32	351	0.351
Create ones / NumPy 64 × 64 · f32	2064	2.064
Create ones / Kernix 64 × 64 · f32	431	0.431
Create ones / faer 64 × 64 · f32	391	0.391